

1 *Research article*

2 **Future inorganic nutrient and plankton dynamics in a temperate lake**

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Abstract

Lakes are experiencing high temperatures and frequent phytoplankton blooms. These events threaten the health of lake ecosystems. Phytoplankton respond to lake warming in various ways, making it challenging to predict changes in community composition. Although understanding the effects of lake warming on phytoplankton communities is crucial for lake management, these effects are not clear and have been a matter of debate due to contrasting observations. Here we use a data-informed, trait-based model to disentangle and project, based on future warming scenarios, the effects of temperature on the dynamics of inorganic nutrients and plankton. Assuming identical thermal characteristics among phytoplankton size classes, we find that nutrient concentrations and phytoplankton mean cell size increase with rising lake temperature. This is driven by the interplay between reduced growth and nutrient uptake by all phytoplankton under sub-optimal temperatures, resulting in higher summer nutrients and intensified zooplankton grazing on smaller phytoplankton cells. The small phytoplankton typically occurring during winter and spring is replaced by large phytoplankton. In addition, peaks of phytoplankton biomass, typically occurring in early summer, emerge earlier, in early spring. The earlier occurrence of these peaks is caused by a shift in the optimal temperature for phytoplankton growth from spring to late winter. Although, as in all studies of this type, our results are based on various modelling approximations, they highlight the importance of combining observations and trait-based modelling for gaining insights into the future compositions of lake phytoplankton and establish a foundation for examining lake ecosystems in a warming world.

Introduction

Most lakes on our planet are under threat by climate change and, specifically, by increasing temperatures (O'Reilly et al. 2015; Woolway et al. 2022). The increase in lake water surface temperature, hereafter water temperature, couples with changes in nutrient conditions in the lake to alter the interactions and compositions of lake communities. This may dampen trophic transfer efficiencies, with serious impacts on the stability and productivity of lake food webs (Strecker et al. 2004; Barneche et al. 2021). Increasing occurrences of anomalous algal blooms (dominated, for example, by cyanobacteria) are observed in eutrophic lakes worldwide, especially in connection to rising temperatures (Paerl and Huisman 2008; Ho et al. 2019). Nevertheless, the mechanistic reasons leading to these blooms are still not clear. In a number of temperate lakes, contrary to expectations (e.g., Pomati et al. 2020), a clear multi-decadal warming trend and a reduction in nutrient levels were linked to the dominance of large phytoplankton cells. Hence, a better understanding of the thermal responses of plankton communities is key to managing lake ecosystems in a warming world.

The effect of temperature on lake phytoplankton communities can be of two types: (1) direct effects can alter metabolic processes of cells and (2) indirect effects can reshape the interactions of biotic and abiotic components (Zohary et al. 2021). Elevated temperatures can indirectly affect phytoplankton nutrient uptake by varying autotrophic metabolic rates and nutrient recycling by varying remineralization rates (Vanni 2002; Dory et al. 2024). Additionally, temperature affects the composition of phytoplankton communities indirectly by enhancing zooplankton grazing, which can favour grazer-resistant phytoplankton species (Sommer and Lewandowska 2011; Lürling 2021). The synergistic direct and indirect effects of temperature can have complex repercussions on phytoplankton communities. These repercussions are frequently observed but they remain difficult to disentangle and predict.

Warming-induced changes in the seasonal succession of phytoplankton have been modelled and observed in different European lakes over the last two decades (reviewed in Winder and Sommer 2012; Vadadi-Fülöp and Hufnagel 2014). Phenological shifts in planktonic communities impact key lake ecosystem functions such as productivity, diversity, and trophic transfer efficiency (Strecker et al. 2004; Woolway et al. 2022). Minimal predator-prey models proposed that earlier onsets can be caused by warming-induced shifts or decoupling of plankton dynamics (Scheffer et al. 2001; De Senerpont Domis et al. 2007a). A competition model based on several phytoplankton species found that warming-induced shifts observed in phytoplankton succession were driven by species-specific responses to temperature change (De Senerpont Domis et al. 2007b). Another way temperature can influence key ecosystem functions is by altering the size composition of phytoplankton, often with opposing effects, especially in the context of temperate lakes. Some mesocosm experiments found that warming favors small phytoplankton cells (Yvon-Durocher et al. 2011; Rasconi et al. 2015), whereas others observed a shift towards large cells (Rüger and Sommer 2012; Yvon-Durocher et al. 2015). Long-term field studies also showed similar contrasting results: high water temperatures fostered both large (Pomati et al. 2020; Dory et al. 2024) and small phytoplankton (Zohary et al. 2017).

The major effects of warming on phytoplankton sizes appear to be driven by temperature-mediated trophic interactions and resource availability (Murphy et al. 2020; Zohary et al. 2021). However, these two drivers are confounded in natural environments and that limits our ability to make sound predictions on the compositions and dynamics of planktonic communities in lake ecosystems. Mathematical models are useful tools for disentangling the interacting effects of temperature on lake plankton. Recent modelling investigations included more comprehensive ecological interactions between plankton communities and the environment. By considering seasonal variations in light, water

temperature, mixed layer depth, and nutrients in a plankton ecosystem model, Cagle and Roelke (2021) and To et al. (2024) found that grazing characteristics of zooplankton regulate the succession of different phytoplankton species and their community composition in temperate lakes. However, such modelling investigations have not been implemented in combination with a high degree of biotic and abiotic data inputs. A data-informed model that is verifiable by in-situ observations can strengthen our understanding on the effects that lake warming can have on composition of phytoplankton communities.

Building on our previous NPZD modelling study (To et al. 2024), we present here an updated data-informed trait-based model to explain plankton dynamics in a specific temperate lake and to explore lake ecosystem responses under different warming scenarios. Greifensee, a medium-size eutrophic lake near Zurich (Switzerland), experienced multi-decadal warming and re-oligotrophic trends (Pomati et al. 2020; Merz et al. 2023). The composition of the phytoplankton community in the lake was found to be sensitive to long-term changes in thermal and nutrient conditions (Bürge et al. 2003; Monchamp et al. 2017), making Greifensee a natural laboratory for studying the impacts of changing temperature on abundance and community composition of lake plankton. This new model version includes additional physiological details on the thermal dependence of the plankton communities (presented in Methods) and assumes that temperature impacts phytoplankton growth and zooplankton grazing through data-derived relationships. The physical processes in the model are driven by observed environmental data and the model outputs are calibrated against averaged nutrient concentration, phytoplankton and zooplankton biomasses, and size data collected at Greifensee. The calibrated model is then run into the future to predict the evolution of the ecosystem under projected climate change scenarios. We considered three Representative Concentration Pathways (RCPs), RCP 2.6, RCP 6.0, and RCP 8.5, reflecting low, medium, and high emission scenarios, respectively. We aim to understand the effect of increased water temperature on future nutrient concentrations,

plankton biomasses, size composition of the phytoplankton community, and timing of the peaks in phytoplankton biomass.

Methods

Size-based plankton ecosystem model

We consider a size-based plankton model that builds on a previous NPZD-based modelling study (To et al. 2024) but that captures now a real lake, Greifensee. The model consists of: (1) an inorganic nutrient, N (i.e., phosphate, PO_4^{3-}), (2) a community of phytoplankton A_i , subdivided into ten size classes ($i = 1, \dots, 10$), covering the phytoplankton size range typically observed at Greifensee, (3) a community of herbivorous zooplankton Z_j subdivided into two size classes ($j = 1, 2$), and (4) a detritus pool, D (Fig. 1). The phytoplankton community exploits nutrients and light for growth, while water temperature affects the growth of phytoplankton and the grazing of zooplankton. Since phosphate is considered the most limiting inorganic nutrient for most lakes (Schindler 1978), all state variables are expressed in units of $\mu\text{M P}$. A detailed model description, including equations and formulation of processes, is reported in Supporting Information S1. In the following, we present the novel features of the model, which include the temperature dependence of phytoplankton growth and zooplankton grazing, the model calibration procedure, and the ecological projections under the different warming scenarios.

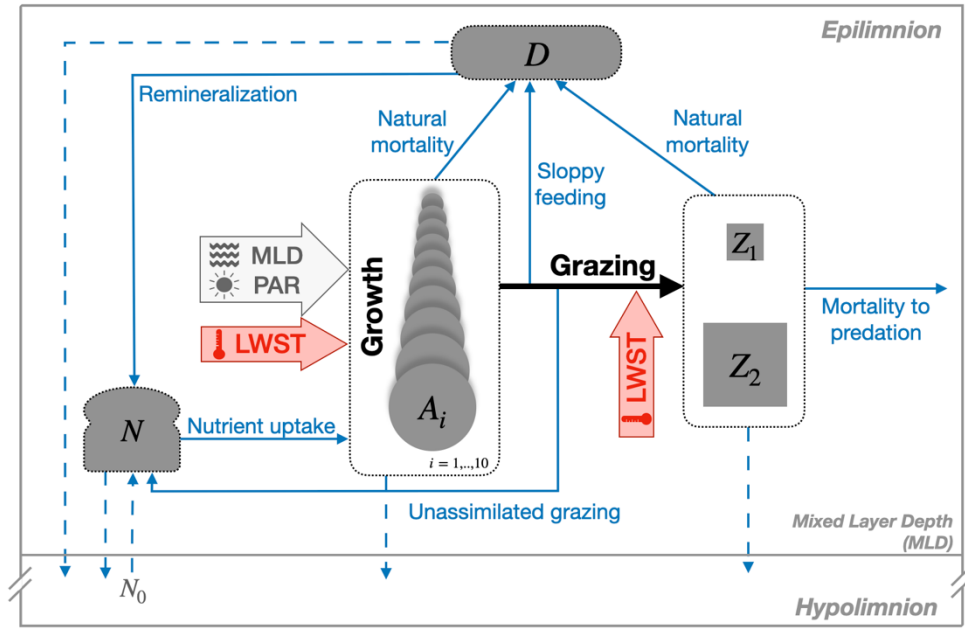


Fig. 1 Schematic of the size-based planktonic model. This model is based on our previous work (To et al. 2024). The new features of this model are highlighted in red. These are the temperature dependences of phytoplankton growth and zooplankton grazing, which are quantified through a unimodal curve and a Q_{10} exponential curve, respectively. The model assumes a zero-dimensional spatial configuration, the so-called slab physics approximation. The epilimnion contains a plankton ecosystem consisting of four components, inorganic nutrient, N , phytoplankton, A_i , zooplankton, Z_j , and detritus, D . Phytoplankton, A_i , is subdivided into i size classes (with $i = 1, \dots, 10$) and zooplankton, Z_j (with $j = 1, 2$), is subdivided into 2 size classes, Z_1 and Z_2 . Phytoplankton growth and zooplankton grazing are size-dependent. The depth of the epilimnion reflects the Mixed Layer Depth (MLD). The blue arrows indicate ecological and biogeochemical processes (solid lines) and physical processes (dashed lines). Environmental forcing to this model is provided by in-situ observations of photosynthetically active radiations (PAR), mixed layer depth (MLD – an indicator of lake mixing), and water temperature (LWST) made at Greifensee. To approximate real-world grazing, the grazed material is split into (1) a portion that is not ingested (due to sloppy feeding) and is immediately lost into detritus, (2) a portion that is ingested and assimilated by zooplankton, and (3) an additional portion that is not assimilated after being ingested and is excreted and quickly remineralised into inorganic nutrients. The loss of zooplankton to higher order predation is density-dependent and formulated as a quadratic function. We assume that the loss of nutrients from the epilimnion is recharged through changes in MLD by a constant nutrient source, N_0 , from the hypolimnion. We determined the value N_0 based on average winter

observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it. The detailed model configuration and all equations describing the processes shown in the schematic are reported in the Supporting Information S1.

Temperature dependences

The basic features of the model considered in this study were introduced by To et al. (2024) and are presented in detail in Supporting Information S1. The novel features include the temperature dependences of phytoplankton growth and zooplankton grazing. The temperature-regulated growth of phytoplankton is described by the following function (Norberg 2004):

$$E(T) = e^{0.063T} \cdot \left[1 - \left(\frac{T - T_{opt}}{\sigma_T} \right)^2 \right], \quad (1)$$

with T representing temperature (°C). T_{opt} and σ_T representing, respectively, thermal optima and thermal tolerance. The curve is set to reflect the observed range of temperatures, between 3 and 23 °C (Eq. 1 and Fig. 2E). Zooplankton grazing is regulated by temperature through the following function (Fig. 2F):

$$\psi(T) = Q_{10}^{\frac{T - T_{ref}}{10}}. \quad (2)$$

The function scales with the Q_{10} coefficient, with T_{ref} representing the reference temperature at which no factorial change in maximum ingestion rate of zooplankton occurs (set to 15 °C, Table 1). The Q_{10} coefficient was determined from a range of values observed by Hansen et al. (1997) for the maximum ingestion rate of Flagellates, Ciliates, and Copepods (functional groups abundant at Greifensee). In our model, the thermal optima (T_{opt}) and the thermal

tolerance (σ_T) for phytoplankton growth and the Q_{10} coefficient for zooplankton grazing are assumed constant. This assumption implies that different phytoplankton and zooplankton size classes have identical thermal responses in relation to, respectively, growth and grazing. These temperature dependencies allow us to examine the effects of temperature on phytoplankton growth and zooplankton grazing and to investigate possible changes to lake ecosystem components when temperature increases.

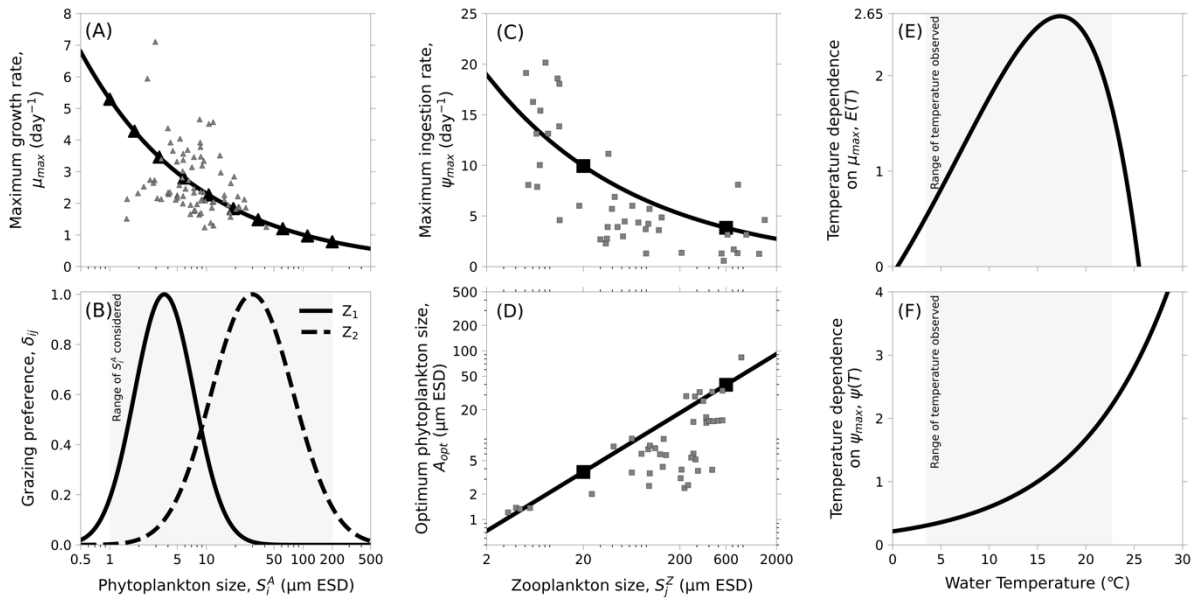


Fig. 2 Model assumptions related to phytoplankton growth and zooplankton grazing. (A) Allometric relationship defining maximum growth rates, μ_{max} , for a given phytoplankton size S_i^A . (B) Size-grazing breadth of the two zooplankton over a range of phytoplankton size classes. Allometric relationships defining (C) maximum ingestion rates, ψ_{max} , and (D) optimal phytoplankton size, A_{opt} , for a given zooplankton size S_j^Z . The community-wide temperature dependence on (E) phytoplankton growth, $E(T)$, and (F) zooplankton grazing, $\psi(T)$. The grey triangles and squares are the data points obtained from literature compilations by Edwards et al. (2012), (A); Hansen et al. (1994), (C); and Hansen et al. (1997), (D). Black triangles and squares represent the specific size values considered in this study for, respectively, phytoplankton and

zooplankton. Shaded areas in panel B represent the range of phytoplankton sizes considered in this study, whereas in panels E and F represent the present observed range of water temperatures at Greifensee.

Model calibration

Environmental forcing

The model is forced with environmental variables observed at Greifensee, a medium-size (about 8.45 km², with a maximum depth of 33 m) eutrophic lake near Zurich (Switzerland). The environmental variables are: water temperature, Photosynthetically Active Radiation (PAR), and Mixed Layer Depth (MLD). Water temperature is obtained from the CTD profiles as daily average epilimnetic temperature from March 2019 to December 2022. The PAR data is converted from the solar irradiance data collected daily by MeteoSwiss, from a weather station next to Greifensee, for the same period as the water temperature. We retrieved data for mixed layer depth from Pomati et al. (2020), which documented monthly values at Greifensee from 1972 to 2015. The forcing functions for the model are daily values over a one-year period. We used daily means from a three-year dataset (March 2019 to December 2022) for water temperature, 75-percentiles from the same three-year dataset for PAR, and daily values obtained by interpolating the monthly averages for MLD (Fig. 3). Details of the procedures followed for generating the forcing functions are reported in Supporting Information S2 and the environmental data are provided in Supplementary Data.

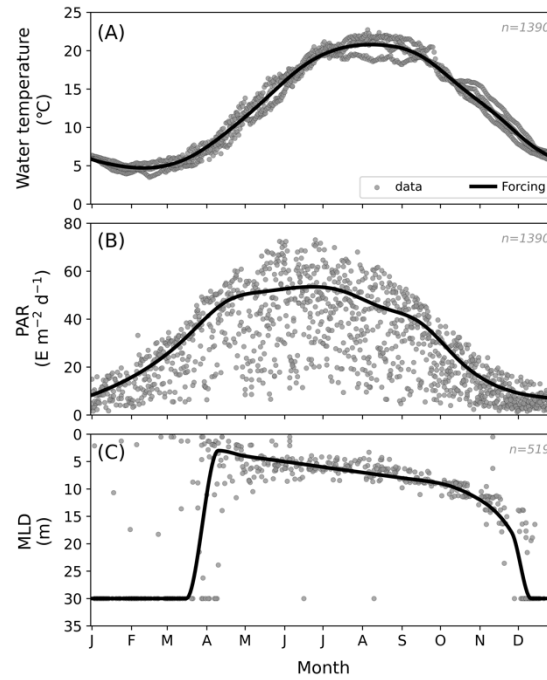


Fig. 3 Forcing functions used in our model. (A) Water temperature ($n = 1390$). (B) Photosynthetically Active Radiation, PAR ($n = 1390$). (C) Mixed Layer Depth, MLD ($n = 519$). A more detailed description of the forcing functions is provided in Supporting Information S2.

Data for model calibration

The model was calibrated by daily mean nutrient concentrations (phosphate) and phytoplankton and zooplankton abundance and size data, provided by the Swiss Federal Institute of Aquatic Science and Technology (Eawag). These data covered the period from March 2019 to December 2022. We selected the six most dominant groups of phytoplankton, namely, chlorophytes, rhodomonas, cryptophyceae, centric diatoms, dinobryon, and uroglena, and two dominant groups of herbivorous zooplankton, namely, ciliates and daphnia. The plankton data were collected at Greifensee using a novel dual-magnification dark-field imaging microscope deployed at 3 meters deep (Merz et al. 2021). The measured abundance data were in units of ROI/sec (Regions Of Interest per second). This pseudo-unit is a statistical output

derived from each detected and processed object by the imaging microscope and is analogous to object counts for both unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). We converted these data into plankton abundances (individual L⁻¹) based on laboratory-calibrated functions (Fig. S1, Table S1), derived by Merz et al. (2021). The plankton abundances were then converted into phosphorus-equivalent biomass concentrations, following previously reported scaling functions based on the size of phytoplankton and zooplankton (Supporting Information S3).

The size data were collected as bio-area (μm²) and converted into biovolume, based on an empirical scaling law (Table S1). The median bio-area of the six phytoplankton species we selected ranged from 10² to 10⁵ μm². Our model captures cell size in terms of equivalent spherical diameter, ESD (μm), because it encompasses differences in size and geometric shape of plankton (Jennings and Paslow 1988). Therefore, in the model, we consider a size spectrum ranging from 1 to 200 μm ESD for the phytoplankton community, encompassing 10 size classes separated by logarithmic intervals. For zooplankton, we considered two size classes, Z₁ and Z₂, of 20 and 600 μm, respectively. We started our simulations with an initial concentration of 6.25e⁻⁴ μM P for all planktonic organisms. We analyzed the phytoplankton size composition in macroecological terms (e.g., Acevedo-Trejos et al. 2018; To et al. 2024), by considering a community-weighted mean cell biovolume, $\overline{V^A}$:

$$\overline{V^A} = \frac{\sum_i^n A_i V_i^A}{\sum_i^n A_i} \quad (3)$$

where n denotes the total number of size classes, and A_i and V_i^A denote, respectively, total biomass and biovolume of the i^{th} class of phytoplankton.

All model parameters were taken from literature (Table 1). Parameters that regulate nutrient uptake and light harvesting of phytoplankton and grazing of zooplankton were fine-tuned manually (i.e., α_{PI} , K_n , $\alpha_{\psi_{max}}$, $\alpha_{A_{opt}}$, $\theta_j^{Z_1}$ and $\theta_j^{Z_2}$; within ranges of values reported in the literature, see Table 1), in order to find best fits (based on visual inspection) between model results and observations (Fig. S5). This parameter calibration produced what we call a ‘standard model run’, i.e. a model run that yields the best match with observations. The time step of our model is one day. To calibrate parameters, we compared the model results against three consecutive years of nutrient and biological data, using daily averages over a one-year period (details can be found in Supporting Information S3). The model results, therefore, reflect the daily dynamics of the lake for a “typical” (i.e., average) year under current temperature conditions. Additionally, we undertook a sensitivity analysis to evaluate the effects of parameter uncertainty on model results. We addressed those parameters ($n=14$) that could not be directly constrained by the data specifically available for Greifensee and those that were not calibrated (Supporting Information S4). Since we are interested in the average state of the lake over one year, the model is run to steady-state for 10 years, using the same annual forcing for every year, and considering only the final year for analysis. Interannual variability was not explored as it was beyond the scope of our study.

Table 1. List of model parameters with corresponding symbols, standard values, parameter range, units, and literature sources. When we report a parameter range, the standard value of the parameter was derived from model calibration.

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Initial slope of P-I curve	α_{P_I}	0.02	0.02 - 0.1	Ein m ⁻² d ⁻¹	Anderson et al. (2015)
Maximum photosynthesis rate	P_{max}	1.1	-	d ⁻¹	To et al. (2024)
Light attenuation coefficient	K_{par}	0.1	0.08 – 0.5	m ⁻¹	Chen et al. (2002); Genkai-Kato et al. (2012)
Thermal tolerance	σ_T	12.5	-	°C	Greifensee data
Thermal optima	T_{opt}	13	-	°C	Greifensee data
Half-saturation constant for nutrient (phosphorus)	K_n	0.06	0.06 -0.1	μM P	Edwards et al. (2012)
Slope of the allometric relationship for maximum growth rate, $\mu_{max_i}(S_i^A)$	$\alpha_{\mu_{max}}$	-0.36	-	-	Edwards et al. (2012)
Intercept of the allometric relationship for maximum growth rate, $\mu_{max_i}(S_i^A)$	$\beta_{\mu_{max}}$	10 ^{0.69}	-	d ⁻¹	Edwards et al. (2012)
Cross-thermocline diffusive mixing coefficient	ω	0.1	-	m day ⁻¹	Fasham et al. (1990)
Natural mortality - Phytoplankton	ϕ_A	0.1	-	d ⁻¹	this study
Natural mortality – Zooplankton	ϕ_Z	0.1	-	d ⁻¹	To et al. (2024)

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Higher-order mortality rate	η_Z	0.34	-	d ⁻¹	Oschlies & Schartau (2005)
Fraction of grazing not egested	ε	0.69	-	-	Fasham et al. (1990)
Fraction of grazing assimilated to growth	γ	0.75	-	-	Oschlies & Schartau (2005)
Remineralization rate	φ	0.15	-	d ⁻¹	Kerimoglu et al. (2017)
Half-saturation constant to phytoplankton cells	K_A	0.19	-	μM P	Kerimoglu et al. (2017)
Size-grazing breadth - Z_1, Z_2	$\theta_j^{Z_1}, \theta_j^{Z_2}$	0.435, 0.6	0.2-0.8	μm ESD	Banas (2011)
Slope of the allometric relationship for maximum ingestion rate, $\psi_{max_j}(S_f^Z)$	$\alpha_{\psi_{max}}$	-0.28	-0.25 - -0.4	-	Hansen et al. (1997)
Intercept of the allometric relationship for the maximum ingestion rate, $\psi_{max_j}(S_f^Z)$	$\beta_{\psi_{max}}$	23	22-26	d ⁻¹	Hansen et al. (1997)
Q ₁₀ coefficient for the maximum ingestion rate	Q ₁₀	2.8	2.0 - 3.9	-	Data in Hansen et al. (1997)
Reference temperature for the Q ₁₀ coefficient	T_{ref}	15	-	°C	Hansen et al. (1997); Kerimoglu et al. (2017)

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Slope of the allometric relationship for optima phytoplankton size, $A_{opt_j}(S_j^Z)$	$\alpha_{A_{opt}}$	0.7	-	-	Hansen et al. (1997)
Intercept of the allometric relationship for optimal phytoplankton size, $A_{opt_j}(S_j^Z)$	$\beta_{A_{opt}}$	0.45		$\mu\text{m ESD}$	Hansen et al. (1997)
Cell sizes – phytoplankton	S_i^A	1-200	-	$\mu\text{m ESD}$	Greifensee data
Body sizes – zooplankton	S_j^Z	20 (Z_1) 600 (Z_2)	-	$\mu\text{m ESD}$	Greifensee data
Initial conditions for nutrient	N_0	0.75	-	$\mu\text{M P}$	Greifensee data
Initial conditions for phytoplankton, zooplankton, and detritus	A_0, Z_0, D_0	6.25e-4	-	$\mu\text{M P}$	this study

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267 Ecological projections based on climate change scenarios

268 The model parameterization obtained from the community-integrated outputs of the standard
269 model run produces model results that reflect current average annual conditions (in terms of
270 nutrient concentrations, total phytoplankton and zooplankton biomasses and aggregated
271 phytoplankton size structures) at Greifensee. With this parameter configuration, we forced the
272 model with new temperatures, representing three climate change scenarios, based on three
273 Representative Concentration Pathways (RCPs), Table S3, expected by the end of this century

(i.e., year 2099) in the region covering Greifensee, between 40° N and 60° N. We considered average and seasonally uniform increases in water temperature by 1.4 °C (RCP 2.6), 2.5 °C (RCP 6.0), and 4.2 °C (RCP 8.5). However, our future projections did not include physical feedback that could be induced by increasing temperatures, like changes in lake stratification (but see Supporting Information S5 for a sensitivity run exploring the impact of this omission on our results under scenario RCP 8.5). Also for these climate change scenarios, we run the model to steady-state for 10 years, using the same annual forcing for every year, and considering only the final year for analyses. In our analyses, annual seasons are defined as follows: winter, from December to February; spring, from March to May; summer, from June to August; and autumn, from September to November.

Sensitivity analyses on phytoplankton thermal traits

For each temperature scenario, we undertook a series of sensitivity analyses on the thermal optima and the thermal tolerance of the phytoplankton community. These sensitivity analyses examine the effects on the model results (annual medians) under variations of $\pm 15\%$ from standard values (Fig. 4). These thermal traits play a key role in regulating phytoplankton responses to temperature change but are difficult to measure in natural systems and their values can vary within and between species (Boyd et al. 2013; Thomas et al. 2016). The range of variability we considered for these sensitivity analyses covers the values of thermal optima and thermal tolerance observed in freshwater species (Thomas et al. 2016).

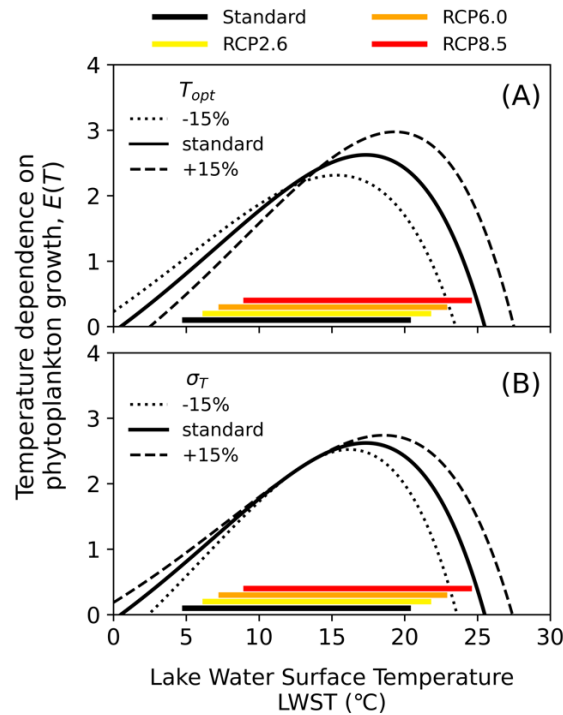


Fig. 4 Temperature dependence curves tested in the sensitivity analyses. These curves result from changes of $\pm 15\%$ in (A) thermal optima, T_{opt} , and (B) thermal tolerance, σ_T , from standard values. The standard values for T_{opt} and σ_T are, respectively, 13 °C and 12.5 °C (black curves). The colored bars represent the annual ranges of water temperatures for various warming scenarios: dark (current average state); yellow (RCP 2.6); orange (RCP 6.0); and red (RCP 8.5).

Results

Current status in the seasonal dynamics of lake ecosystem components

For the model calibration, we focused on the seasonal dynamics of four major variables: (1) nutrient concentration, (2) phytoplankton biomass, (3) zooplankton biomass, and (4) phytoplankton mean biovolume (the latter three are community-integrated). The simulated inorganic nutrient dynamics capture the average annual cycle of epilimnetic phosphorus observed at Greifensee (Fig. 5A). The nutrient concentration is characterized by a steady

decline, starting at the beginning of the year and lasting for about three months, it remains low throughout summer and for part of autumn, and it restocks by the end of the year (Fig. 5A).

The simulated plankton dynamics also capture the observed community-integrated biomass qualitatively well (Fig. 5B). The model captures the biomass accumulation of phytoplankton during spring, summer, and autumn, and reproduces correctly the low phytoplankton biomass levels after the summer and autumn peaks (Fig. 5B). However, the model produces a first winter peak that is not shown by the data (Fig. 5B), it underestimates the spring biomass, and it overestimates the summer biomass. The simulated zooplankton biomass matches well with the autumn and winter observations, despite a slight overestimation for the spring peak (Fig. 5C). A mismatch is noticeable in June and July, when the simulated zooplankton biomass declines due to the dominance of large and less grazable phytoplankton (Fig. 5C and 5D). In summary, despite being an approximation of the real lake in terms of physics and ecology, the model captures several general features of the phytoplankton and zooplankton seasonal cycles; namely the spring, summer, and autumn biomass accumulations of phytoplankton and the spring biomass peaks of zooplankton.

A good match is also obtained between the simulated and observed seasonal patterns in the community-mean biovolume of phytoplankton. In general, both model and data show small mean biovolume in late winter and beginning of spring (Fig. 5D), when smaller cells outcompete larger cells due to weaker grazing on smaller cells (Fig. 5C). Relatively larger phytoplankton cells are present during summer and autumn (Fig. 5D), when zooplankton grazing on smaller cells increases and the larger, less grazable cells gain dominance. Additionally, we find that the phytoplankton size composition is strongly controlled by grazing characteristics, namely, the allometric scaling of grazing (i.e., maximum ingestion rate and

optimal phytoplankton size) and the size-grazing breadth of zooplankton (Fig. S3A, S5I, S5J, S5L).

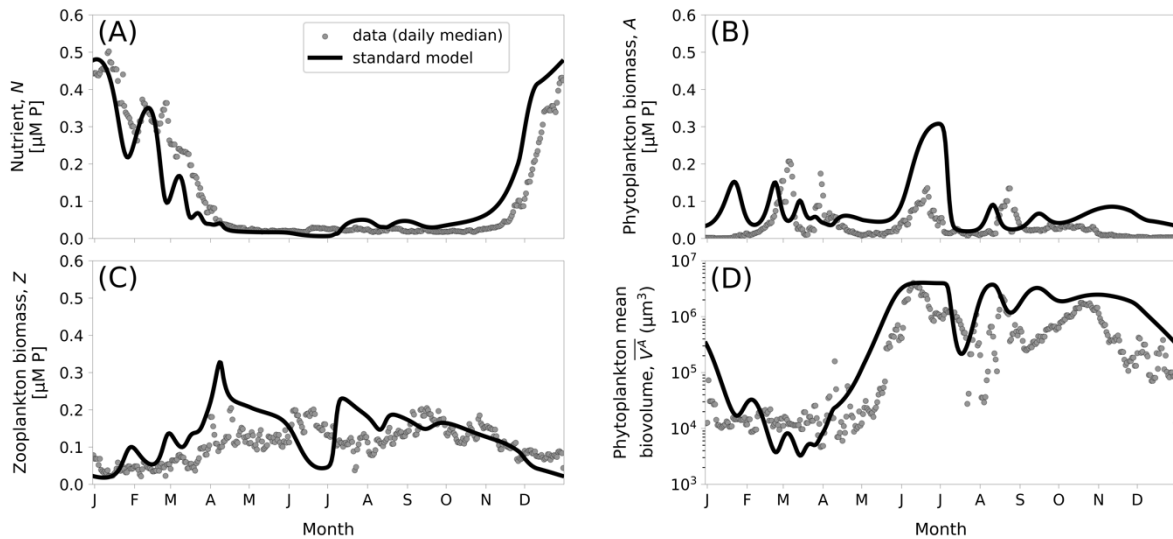


Fig. 5 Results of the standard run showing temporal dynamics of (A) nutrient concentration, (B) phytoplankton community biomass concentration, (C) zooplankton community biomass concentration, and (D) phytoplankton mean biovolume. The grey dots represent the observational data collected in the epilimnion by the underwater imaging device (see details about data transformation in Supporting Information S3) and the black lines represent the model outputs. The model is run to steady-state over 10 years and we show here the results of the 10th year. The observations are median daily values over 3 years of data (from March 2019 to December 2022).

Expected future seasonal dynamics of major variables

Alterations in nutrient dynamics

We find seasonal alterations in the nutrient dynamics under the warming scenarios, with the strongest alteration occurring under RCP 8.5 (water temperature higher by 4.2 °C with respect

to current average). The most prominent change in the nutrient concentration occurs in summer and autumn, i.e. from July to October, when there are excessive nutrient accumulations in the lake, and accumulation intensifies with the warming scenarios (Fig. 6B). Under RCP 8.5, the nutrient accumulation in summer reaches an annual maximum, a pattern that contrasts sharply with the current status (standard run) when the highest nutrient level occurs in winter (Fig. 6B). This summer nutrient peak coincides with a decline in phytoplankton biomass, caused by the unfavorable temperature conditions for growth experienced by the system under RCP 8.5 (Fig. S6A and S7). Warming also causes an increase in the overall annual nutrient concentration in the epilimnion. Under RCP 8.5, the annual median nutrient concentration is about four times higher than that under the current status, i.e. $0.05 \mu\text{M P}$ (current status) versus $0.2 \mu\text{M P}$ (RCP 8.5), Fig. 6C. In summary, our model projections suggest that as the lake warms up, nutrient accumulation increases in the summer months and the lake becomes more eutrophic.

Alterations in plankton phenology

As the lake warms up, the model shows lower winter peaks and earlier summer peaks of phytoplankton (Fig. 6D). The timing of the summer peaks depends on the timing of thermal optima, which shifts with the warming scenarios (Fig. S6C). Under RCP 8.5, the summer phytoplankton peak advances by about two months, from July to May. The earlier arrival of the summer peaks shortens the corresponding period of low phytoplankton biomass after the spring peaks. We also notice an increased number of minor phytoplankton peaks during the year as warming intensifies, with small reductions in annual median phytoplankton biomass concentrations (Fig. 6E). In summary, the increase in water temperature advances the timing of the major phytoplankton peaks and increases the number of minor peaks occurring throughout the year.

The model results also show phenological shifts for zooplankton. The timing of the first zooplankton peak in April remains the same in all climate scenarios, but the second peak moves earlier with the severity of the RCP scenarios (Fig. 6F). Consequently, the period between the first and the second zooplankton peaks shortens by two months under RCP 8.5. Also, the zooplankton biomass in late summer decreases noticeably under RCP 8.5, compared to other scenarios, due to the decline in phytoplankton growth (Fig. S7). The annual median zooplankton biomass is lower in RCP 8.5 compared to other scenarios (Fig. 6G). In summary, major alterations in the phenology of zooplankton include a reduction in the period between the first and the second peak and a reduction in the summer biomass.

Alterations in phytoplankton mean biovolume

An elevated water temperature increases the mean biovolume of phytoplankton and reduces the range of size variations during the year. In particular, as the lake warms up, relatively smaller cells are replaced by larger cells in winter and in spring (Fig. 6H). Under RCP 8.5, the dominance of relatively larger phytoplankton appears already in April, two months earlier than in the current status. In summary, a warmer lake fosters a narrower size diversity of phytoplankton and reduces seasonal variations in mean biovolumes (Fig. 6I).

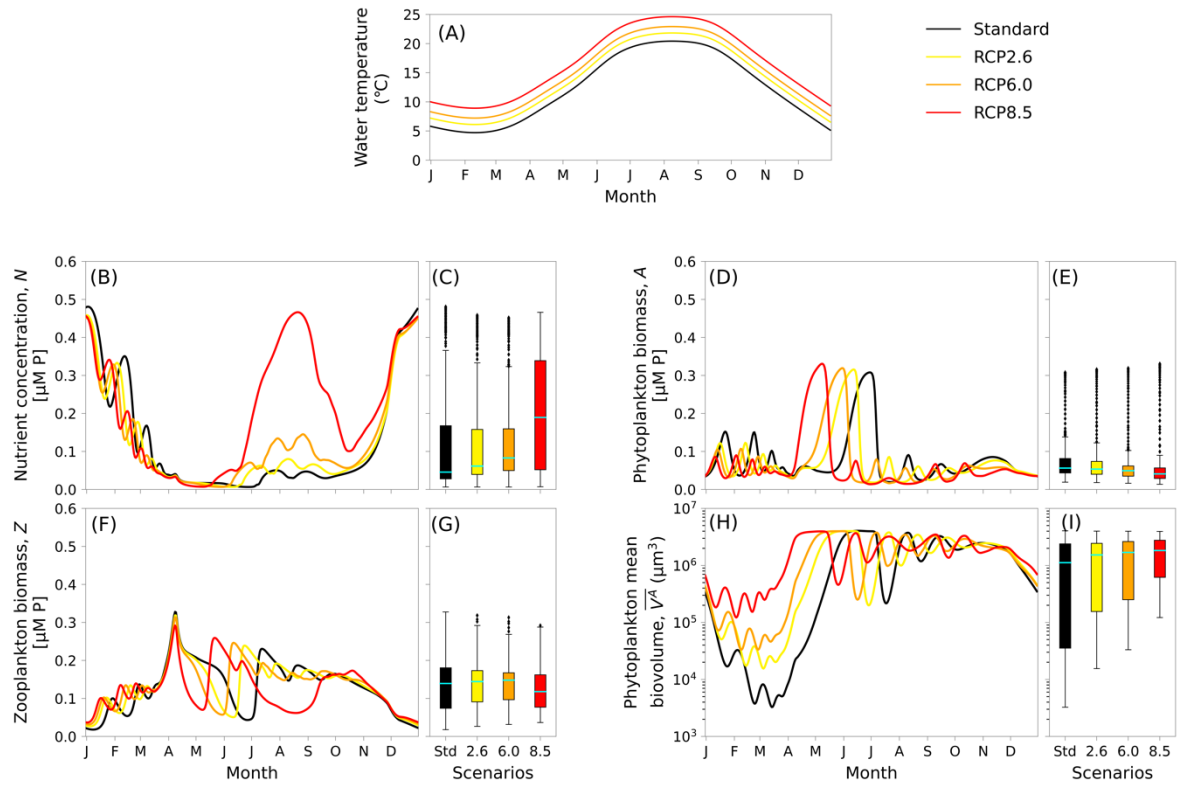


Fig. 6 Results of climate change scenarios compared to the current average status of the lake (standard run, black lines and Fig. 5). The climate change scenarios, RCP 2.6, RCP 6.0, and RCP 8.5 simulate an average and seasonally uniform increase in water temperature by, respectively, 1.4 °C, 2.5 °C, and 4.2 °C (A). The results show temporal dynamics of (B) nutrient concentration in the epilimnion, (D) phytoplankton biomass, (F) zooplankton biomass, and (H) phytoplankton mean biovolume. Boxplots (panels C, E, G, and I) show the overall seasonal variability (width) and median (cyan lines) of the corresponding variables. The model is run to a steady annual cycle over a period of 10 years with repeated annual forcing, the results shown are those of the 10th year.

Sensitivity analysis of future scenarios

We examined the sensitivity of the model results by varying the values of the thermal traits, thermal optima (T_{opt}) and thermal tolerance (σ_T), by $\pm 15\%$ from standard values. The results show that, under future warming scenarios, the nutrient concentration, among other variables,

is most sensitive to changes in the thermal traits. A reduction in either of the thermal traits leads to a higher median nutrient concentration, whereas an increase in either of the thermal traits leads to a lower median nutrient concentration (Fig. 7). These changes are exacerbated by the severity of the warming scenarios. Under RCP 8.5 (Fig. 7C), the median nutrient concentration increases by at least 200% when the phytoplankton community is characterized by lower values of thermal traits (i.e., lower tolerance to higher temperatures, Fig. 4). Such higher nutrient concentrations are caused by reduced growth of phytoplankton, from spring to autumn, due to a less suitable temperature for growth (Fig. S8G, S8I). Additionally, lower T_{opt} or σ_T leads to pronounced reductions in phytoplankton and zooplankton biomasses and in phytoplankton mean biovolume, particularly under RCP 8.5 (Fig. 7C). A higher T_{opt} (or σ_T) does not induce as many variations to the model variables; even under RCP 8.5, the greatest relative change is less than 50%. That is because higher thermal trait values produce a more heat-tolerant phytoplankton community (Fig. 4).

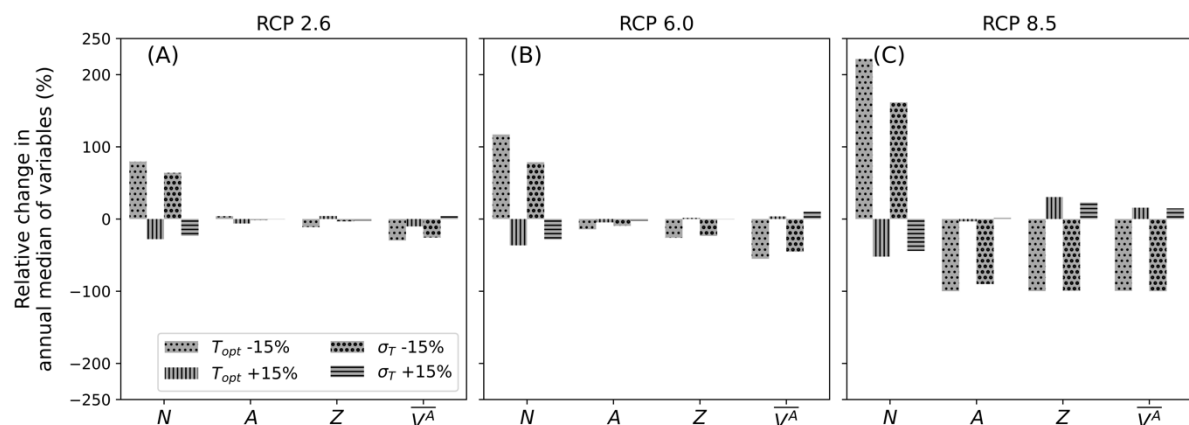


Fig. 7 Sensitivity of model variables (annual medians) caused by $\pm 15\%$ changes from standard values of thermal optima (T_{opt}) or thermal tolerance (σ_T), for the different climate change scenarios: (A) RCP 2.6, (B)

417 RCP 6.0, and (C) RCP 8.5. The model variables are nutrient concentration, N ; phytoplankton biomass, A ;
418 zooplankton biomass, Z ; and phytoplankton mean biovolume, $\overline{V^A}$.

419

420 **Discussion**

421 Mounting evidence over the last two decades shows rising lake temperatures on a planetary
422 scale, with European temperate lakes expected to experience the fastest warming trend
423 (Woolway et al. 2022). The consequences of rising temperatures on the physiology and ecology
424 of plankton are far from clear. In our model, the growth of phytoplankton depends, among
425 other factors, on a thermal dependence curve characterized by data-constrained (specifically,
426 Greifensee data) optimal temperature for growth and tolerance range; and the grazing of
427 zooplankton depends on a Q_{10} function constrained by literature data. Under future climate
428 change scenarios, our model predicts increased epilimnetic nutrient concentrations,
429 advancements in phytoplankton biomass peaks, and dominance of relatively larger
430 phytoplankton. We also find that a phytoplankton community with lower thermal optima or
431 lower thermal tolerances leads to even higher accumulation of nutrient concentrations due to
432 the less favourable temperature ranges of phytoplankton growth and thus of lower nutrient
433 uptake.

434 *Current seasonal patterns of nutrients and phytoplankton concentrations*

435 With respect to the current average state of Greifensee, the seasonal pattern of phosphorus
436 dynamics produced by the model matches well with the observational data and reflects the
437 typical nutrient dynamics of temperate lakes, which are characterized by high nutrient
438 concentrations in winter and low concentrations in summer (Sommer et al. 2012). In temperate
439 lakes, relatively abundant phosphorus at the beginning and at the end of the year coincides with

scarce light and less favourable temperatures for growth, resulting in lower growth and uptake rates (Staehr and Sand-Jensen 2007; Sommer et al. 2012). The June-July peaks in the observed phytoplankton and zooplankton biomasses are temporally coincident, whereas the model results show a slight shift, with zooplankton peaking slightly later than phytoplankton. This small delay, typical of predator-prey models, is an expected feature of our model, given its general resource-prey-predator structure. We find that the annual variations in the mean phytoplankton biovolume are characterized by relatively higher abundances of smaller phytoplankton in spring, higher abundances of larger phytoplankton in summer, and higher abundances of intermediate-size phytoplankton in autumn. These patterns are consistent with the general phytoplankton dynamics expected to occur when interactions between inorganic nutrient limitation and grazing pressure are primary driving factors of lake ecosystems (Vanni and Temte 1990; Sommer et al. 2012). Our model further supports findings from previous size-based models (Banas 2011; Taniguchi et al. 2023; To et al. 2024) showing that the biomass of phytoplankton is regulated by variations in resources like light and nutrients, whereas the annual variations in the phytoplankton size composition are strongly controlled by grazing characteristics like the size-feeding breadth of zooplankton. By using appropriate zooplankton size-feeding characteristics under the influence of major physical factors in lakes, we capture and elucidate the phenology of phytoplankton biomass and size composition at Greifensee.

Future increased phosphorus concentration driven by phenological shift of phytoplankton

The ecological projections under future warming scenarios show that epilimnetic phosphorus concentrations increase with the increasing severity of global warming. The higher phosphorus concentration is driven by a shift in phytoplankton phenology, characterized by lower accumulations of phytoplankton biomass in winter and earlier accumulations in spring, causing a reduction of phytoplankton biomass in summer due to sub-optimal temperature conditions

for phytoplankton growth. The sub-optimal condition for growth also results from a mismatch between optimal temperature and light conditions, as the optimal temperature for growth occurs earlier than optimal light conditions. This mismatch retains unused phosphorus in the summer epilimnion because of reduced phosphorus uptake by the phytoplankton community. This result is based on the assumption that different phytoplankton size classes have identical thermal characteristics (Fig. 2). In nature, species sensitivities to temperature may vary and different phytoplankton can exhibit different growth and nutrient uptake abilities along thermal gradients (Thomas et al. 2016; Bestion et al. 2018). This is an aspect that could generate feedbacks on the nutrient dynamics that are not captured by our model. Also, higher temperatures lead to stronger lake stratification, which in turn has an impact on competition for nutrients and light with consequences on the composition and dynamics of phytoplankton communities (Tilzer and Goldman 1978; Donis et al. 2021). The stronger and longer stratification events in a warmer lake (Butcher et al. 2015; Woolway et al. 2021) can further disrupt the nutrient exchanges between epilimnion and hypolimnion and, thus, influence nutrient dynamics (Supporting Information S5, Fig. S4B). Taken together, these results indicate that accounting for the impacts of climate on nutrient regimes would require detailed considerations in relation to physical and biological feedbacks. Our results should be considered as a first order approximation with respect to the impacts that global warming can have on nutrient dynamics and plankton phenology.

Earlier accumulation of phytoplankton biomass driven by thermal optima

Our climate change scenarios produce earlier accumulations of summer phytoplankton biomass. Earlier formation of spring blooms in temperate lakes was detected by previous studies based on simulations and observations. For example, a dynamic diatom-daphnia model showed that, under lake warming, a high phosphorus supply causes earlier diatom spring

blooms (Huber et al. 2008). A depth-resolved hydrodynamic model coupled to a simple phytoplankton growth model showed that earlier spring blooms can be caused by changes in lake stratification patterns (Peeters et al. 2007). In our model, the earlier peaks of large phytoplankton are produced by a warmer winter that advances the timing of optimal temperature for the growth of all phytoplankton size classes but intensifies the grazing of zooplankton on small phytoplankton size classes. Thermal characteristics, like optimal temperature for growth, are important factors for driving seasonal succession in culture experiments comprising marine diatoms (Anderson and Rynearson 2020). Although our projections are largely consistent with these studies, a long-term observation conducted in a large Norwegian lake revealed delayed phytoplankton peaks (Moe et al. 2022). Their observation considered taxonomic details of the phytoplankton community and showed that the phenological changes of phytoplankton are contingent to the different succession patterns exhibited by individual taxa. Therefore, taxonomic details related to the different thermal responses in a community are important for understanding specific changes in the phenological patterns of phytoplankton (Thackeray et al. 2008; Segura et al. 2018). This shows the importance of considering size or taxonomic differences in thermal traits for capturing specific phenological trends.

Changes in phytoplankton mean biovolume under increasing temperature

We find an increase in mean phytoplankton biovolume and a decrease in the range of variability with increasing average temperature. These trends are caused by the disappearance of smaller phytoplankton during spring and the earlier appearance of larger phytoplankton during summer. Our results are driven by thermally-regulated bottom-up (nutrient uptake by phytoplankton) and top-down (zooplankton grazing) processes. As temperature increases, the optimal temperature for phytoplankton growth occurs earlier and the summer temperatures

become sub-optimal for growth (Fig. S6C); whereas zooplankton grazing, which targets smaller phytoplankton, intensifies (Fig. S6B). These aspects interact not only to favor phytoplankton cells with intermediate biovolumes over cells with small biovolumes, but also to foster the earlier appearance of large phytoplankton. Intensified grazing on smaller phytoplankton was observed in warmer freshwater environments (Winder and Sommer 2012). Analyses of time-series from eight Swiss lakes revealed a rising proportion of larger phytoplankton cells in association with higher temperatures, higher zooplankton abundances, and lower nutrient levels (Pomati et al. 2020). This multi-decadal dataset showed that larger phytoplankton are found in summer, coinciding with higher grazing pressures compared to winter, suggesting the presence of a grazing-mediated positive correlation between larger phytoplankton and higher temperatures. However, in lakes where zooplankton grazing is weak, there can be different results. In Lake Kinneret (Israel), the prevalence of small phytoplankton in stratified summer waters (based on an eight-year observation) resulted from a strong selection by sinking processes on the better-suspended small phytoplankton (Zohary et al. 2017). Sinking was not included in our model because Greifensee is characterized by a relatively shallow average depth (18 meters), but this loss process can contribute to varying the size composition of phytoplankton communities (Zohary et al. 2017). In summary, the physical and biological properties of a lake are important for understanding temperature-induced changes in the community composition of phytoplankton. Our findings here emphasize the role played by intensified grazing on the community compositions of phytoplankton in a warmer Greifensee.

Thermal traits affect future nutrient concentrations

The responses of phytoplankton to increasing temperatures are reflected in traits like optimal temperature for growth and thermal tolerance. Our sensitivity analyses suggest that the future

projections of nutrient dynamics in the lake depend highly on the optimal temperature for growth and thermal tolerance of phytoplankton, which control their growth and uptake efficiencies under different temperature conditions. Previous modelling studies also showed that the thermal traits of phytoplankton regulate community responses. An ecosystem model resolving biogeochemical processes, for example, showed that alterations of the Q_{10} factor can markedly change the community composition of phytoplankton under the most severe climate change scenario (Anderson et al. 2024). An NPZ model driven by seasonal variations of temperature showed that tackling species differences, especially in terms of thermal optima, is important for estimating the thermal sensitivities of phytoplankton (Chen 2022). In addition, optimal temperatures for phytoplankton growth may co-vary with nutrient and light availabilities (Edwards et al. 2016; Bestion et al. 2018) in ways that are not captured by our model. We suggest that measuring size-specific thermal trait values would be a crucial asset for modelling studies such as this one. These data would facilitate the formalization of a robust functional relationship between thermal traits and plankton sizes, a fundamental assumption for investigating the effects of temperature on plankton communities and the corresponding ecosystem feedback under climate change scenarios.

Limitations and future directions

Our study investigates the impact of increased water temperature in a specific lake system, relying on specific, data-derived assumptions regarding temperature-dependent phytoplankton growth and zooplankton grazing. Our results are based on a relatively idealized lake physics, which does not specify vertical stratification and lateral transport processes. We consider that this choice is appropriate for Greifensee, a relatively shallow lake that is typically fully mixed in winter and stratified into two distinct and homogeneously mixed layers in summer (Mieleitner et al. 2008; Huntscha et al. 2018). Many studies (e.g., Livingstone 2003; Rempfer et al. 2010;

Butcher et al. 2015; Woolway et al. 2021) also show that, in a warming world, lakes will experience weaker mixing, with higher temperatures potentially structuring the water column into two stable layers, thus, making our simplified physical scheme an appropriate first-order approximation approach. Also, our results are based on a specific set of model parameters that are informed by lake data. This reduces the level of uncertainty in these parameters, especially those that regulate phytoplankton nutrient uptake and zooplankton grazing (Fig. S3), which are the major determinants of our results. Additionally, previous data suggested temperature dependences that were not considered in this study like phytoplankton respiration (Robarts and Zohary 1987) and nutrient remineralization rates (Boscolo-Galazzo et al. 2018). The absence of size-specific temperature dependencies in our model prevents niche differentiation with respect to thermal traits (Chen 2022). This limits the model ability to maintain all 10 phytoplankton size classes over time due to competitive exclusion (Hardin 1960), an aspect common to plankton models (e.g., To et al. 2025) but that does not have an equivalent in nature (the “paradox of the plankton”, Hutchinson 1961). Thus, some of the size classes simulated by our model disappear during the course of the simulations. This, however, does not preclude the model from capturing, both in magnitude and dynamics, nutrient and plankton concentrations, as well as mean phytoplankton biovolume.

Assessing how these limitations impact the community compositions of phytoplankton would be an interesting avenue for future research. Data-derived size-scaling relationships about optimal temperature for growth or thermal tolerance, which we consider crucial to phytoplankton physiology, remain scant. Resolving these scaling relationships can optimize the mechanistic investigations of size-based models under future scenarios (Chen 2022; Litchman 2023), particularly those in relation to the synergistic effects of temperature, nutrients, and zooplankton grazing on phytoplankton communities (Pomati et al. 2020; Dory et al. 2024). Future modelling studies could examine the impacts of temperature on planktonic

interactions more realistically by considering size-specific thermal responses (e.g., Chen 2022; Karakuş et al. 2022) or other size-related characteristics, like a more diverse zooplankton community in relation to body size. One could also consider additional temperature dependences on, for example, remineralization processes (Boscolo-Galazzo et al. 2018). Assuming that metabolic and biochemical rates would be enhanced by increasing temperatures, these dependences would produce more accurate dynamics of inorganic nutrients and detritus. Spatially explicit hydrodynamic modelling studies could investigate the extent to which our first-order slab physics approximation is appropriate to capture the future ecological regimes of lakes like Greifensee (e.g., Huntscha et al. 2018). Studying inter-annual variabilities would be another obvious next step when looking at future projections of lake ecosystem.

Conclusions

By combining a broad range of field observations and data-informed trait-based modelling, our study provides insights into the future dynamics of nutrient, plankton biomasses and community compositions of phytoplankton in a temperate lake under global warming scenarios. Our model predicts three potential consequences in a warmer lake, namely, increased epilimnetic nutrient retention, advancement of phytoplankton blooms, and increased mean phytoplankton biovolume, the latter being brought about by a replacement of small phytoplankton cells with large ones. Acknowledging a number of simplifications and limitations, this study represents a first-order approximation of the ecological responses that Greifensee may exhibit in a warmer world. Most of the model parameters and all environmental forcing are site-specific, thus the validity of our results is limited to Greifensee or to lake systems with characteristics similar to Greifensee. Future investigations of these phenomena, based on alternative sets of assumptions or expanded datasets, would be a valuable effort to verify the likelihood of their occurrence. Our approach emphasizes the multifaceted responses

of planktonic communities to global warming, highlighting both challenges and opportunities involved in understanding and managing a lake ecosystem in a changing world. We thus argue that long-term monitoring programs, like those undertaken at Greifensee, in combination with modelling studies based on temperature dependences on the physiology and ecology of plankton can improve our predictions and enhance our capacity to design effective lake management strategies. Finally, investigating how the changes predicted by our model could impact lake ecosystem functions would help the scientific community to assess the significance that these studies have for lake management and conservation strategies in a warming world.

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855 **Data Availability Statement**

856 The lake data and model outputs used for this study are available on Zenodo (doi:
857 10.5281/zenodo.11390345). The code to reproduce the analyses of this study is available on
858 Github (<https://github.com/systemsecologygroup/TempSizeMod>).

859

860 **Conflict of Interest**

861 None declared.